Aprendizaje de Máquina

U2 Análisis de clústeres

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Section 1

Sequential Clustering Algorithm

Introduction

The **sequential algorithms** are quite **straightforward** and **fast** methods that produce a **single clustering**.

Characteristics:

- In most of them, all the feature vectors are **presented** to the algorithm **only once**. So, the final result is dependent on the **order** in which the vectors are presented.
- These schemes tend to produce **compact** and **hyperspherically** or **hyperellipsoidally** shaped clusters, depending on the distance metric used.
- The **number of clusters** is **not known** a priori in this case. In fact, new clusters are created as the algorithm evolves.

Introduction

Notation:

Let $d(x, C)$ denote the distance (or dissimilarity) between a feature vector x and a cluster C.

The **user-defined parameters** (hyper parameters) required by the algorithm scheme are:

- The **threshold of dissimilarity** θ.
- The **maximum** allowable **number of clusters** q.

Section 2

Basic Sequential Algorithmic Scheme (BSAS)

Basic idea

The **basic idea** of the algorithm is the following:

As **each new vector** is considered, it is **assigned** either to

- An **existing** cluster.
- A **newly created** cluster

depending on its distance from the already formed ones.

Basic Sequential Algorithmic Scheme (BSAS)

Let m be the number of clusters that the algorithm has created up to now.

Basic Sequential Algorithmic Scheme (BSAS):

1. $m = 1$
2. $C_m = \{x_1\}$
3. For $i = 2, \ldots, N$:
    1. Find $C_k : \; d(x_i, C_k) = \min_{1 \leq j \leq m} d(x_i, C_j)$
    2. If $d(x_i, C_k) > \theta$ and $m < q$ then
        - $m = m + 1$
        - $C_m = \{x_i\}$
    3. Else
        - $C_k = C_k \cup \{x_i\}$
        - Where necessary, update representatives.
    4. End-if
4. End-for

Notes

- The **order** in which the vectors are presented to the BSAS plays an important role in the clustering results. Different **presentation ordering** may lead to totally **different clustering results**, in terms of the number clusters as well as the clusters themselves.
- The **choice of the threshold** affects the **number of clusters formed** by BSAS.
  - If θ is **too small**, unnecessary clusters will be created.
  - If θ is **too large** a smaller than appropriate number of clusters will be created.
- If the number q of the **maximum** allowable **number of clusters** is not constrained, we leave it to the algorithm to *decide* about the appropriate number of clusters.
  - However, **constraining** q becomes necessary when dealing with implementations where the available **computational resources** are limited.

Remarks

- Different choices of **distance measures** $d(x, C)$ lead to different algorithms.
- When C is represented by a representative point m_C, $d(x, C)$ becomes

$$d(x, C) = d(x, m_C)$$

- The BSAS scheme may be used with **similarity** instead of dissimilarity measures with appropriate **modification**; that is, the min operator is replaced by **max** (step 3.1).
- BSAS, using point cluster representatives, tends to favor **compact clusters**, making it unsuitable when there is strong evidence of other cluster types.

Computational complexity

The **BSAS algorithm** performs a **single pass** on the entire data set, X.

For each iteration, the distance of the vector currently considered from each of the clusters defined so far is computed.

Because the final number of clusters m is expected to be much smaller than N, the **time complexity** of BSAS is $O(N)$.

Estimation of the number of clusters

A simple method for determining the **number of clusters** is the following:

- For $\theta = a$ to b step c.[1]
  - Runs s times the algorithm $BSAS(\theta)$, each time presenting the data in **different order**.
  - Estimate the number of clusters $m(\theta)$ as the **most frequent** number resulting from the s runs of $BSAS(\theta)$
- Next θ.

where $BSAS(\theta)$ denotes the BSAS algorithm with a specific threshold of dissimilarity θ.

[1] The values a and b are the minimum and maximum dissimilarity levels among all pairs of vectors in X, that is,

$$\begin{aligned} a &= \min_{i,j=1,\ldots,N} d(x_i, x_j) \\ b &= \max_{i,j=1,\ldots,N} d(x_i, x_j) \end{aligned}$$

Estimation of the number of clusters

In the sequel we plot $m(\theta)$ versus θ. This plot has a number of flat regions.

We estimate the number of clusters as the number that corresponds to the **widest flat region**.[2]

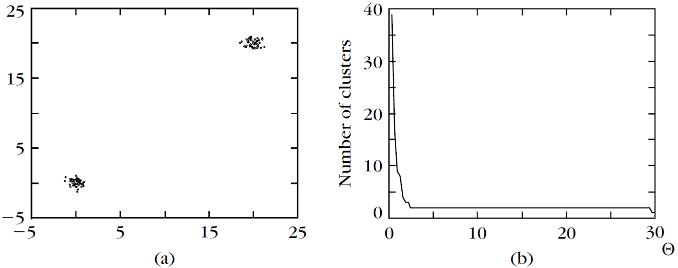
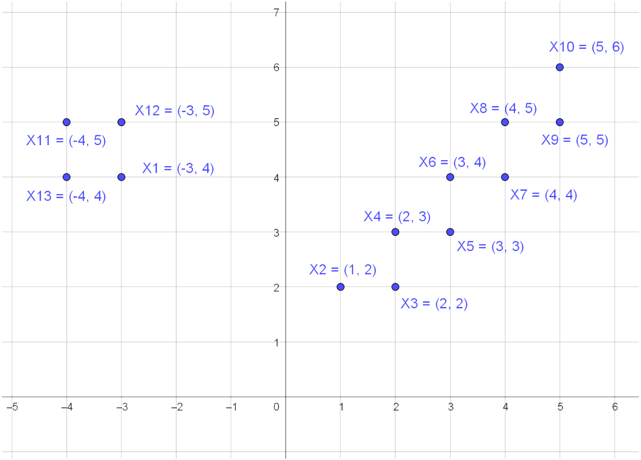


Figure: a) The data set. (b) The plot of the number of clusters $m(\theta)$ versus *theta*.

[2] In this procedure, we have implicitly assumed that the feature vectors do form clusters. If this is not the case, the method is useless.

Exercise

Consider the following two-dimensional vectors and run the BSAS algorithm for different distances d and threshold θ. Try different orders in presenting data:



Section 3

Modified MBSAS

Modified BSAS (MBSAS)

The BSAS take a decision for the vector x **prior to** the final cluster formation, which is determined after all vectors have been presented.

The following refinement of BSAS, which will be called **modified BSAS (MBSAS)**, overcomes this drawback.

The cost we pay for it is that the vectors of X have to be **presented twice** to the algorithm.

Basic idea

The algorithmic scheme consists of **two phases**:

- The **first phase** involves the **determination of the clusters**, via the **assignment** of **some** of the vectors of X to them.
- During the **second phase**, the **unassigned vectors** are presented for a second time to the algorithm and are **assigned** to the appropriate cluster.

Modified BSAS

Modified BSAS (MBSAS):

Cluster determination (first step):

- $m = 1$
- $C_m = \{x_1\}$
- For $i = 2, \ldots, N$:
  - Find C_K : $d(x_i, C_k) = \min_{1 \leq j \leq m} d(x_i, C_j)$
  - If $d(x_i, C_k) > \theta$ and $m < q$ then
    - $m = m + 1$
    - $C_m = \{x_i\}$
  - End-if
- End-for

Modified BSAS

Patter classification (second step):

- For $i = 1, \ldots, N$:
  - If x_i has not been assigned to a cluster, then
    - Find $C_k : \; d(x_i, C_k) = \min_{1 \leq j \leq m} d(x_i, C_j)$
    - $C_k = C_k \cup \{x_i\}$
    - Where necessary, update representatives.
  - End-if
- End-for

The **number of clusters** is determined in the **first phase**, and then it is frozen.

Thus, the decision taken during the second phase for each vector takes into account all clusters.

Iterative mean vector

When C is represented by a single vector $d(x, C)$ becomes

$$d(x, C) = d(x, m_R)$$

where m_C is the representative of C.

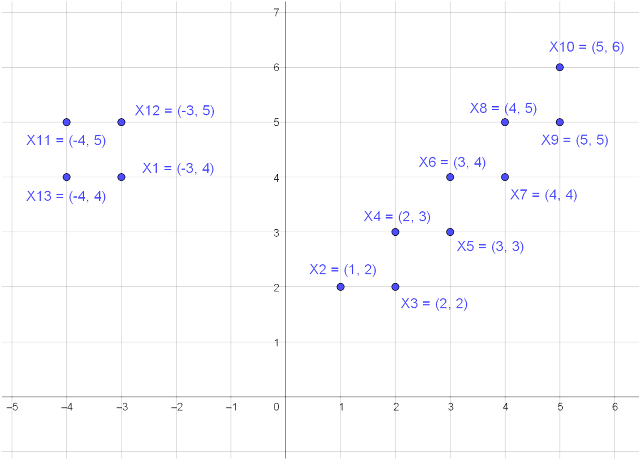
In the case in which the **mean vector** is used as a representative point, the updating may take place in an iterative fashion, that is,

$$m_{C_k}^{new} = \frac{(n_{C_k}^{new} - 1)m_{C_k}^{old} + x}{n_{C_k^{new}}}$$

where $n_{C_k}^{new}$ is the cardinality of C_k after the assignment of x to it and $m_{C_k}^{new}$ and $m_{C_k}^{old}$ are the representative of C_k after and before the assignment of x to it, respectively.

Exercise

Consider the following two-dimensional vectors and run the MBSAS algorithm for different distances d and threshold θ. Compare with the results of the BSAS.



References

- Theodoridis, S. & Koutroumbas, K. *Pattern Recognition*, 4th ed., Academic Press, 2009